List 1 - Funções e Relações

Maria Eduarda Salmo 26704708

1) $A=\{0,2,4,6\}$ $B=\{1,3,5,7\}$

$A \times B = \{(0,1)(0,3)(0,5)(0,7),$
$(2,1)(2,3)(2,5)(2,7),$
$(4,7)(4,3)(4,5)(4,x)$
$(6,1)(6,3)(6,5)(6,7),$

a) $(2,5)$, pois $2 \in A$ e $5 \in B$ → (V)
b) $(3,1)$, pois $3 \notin A$ e $1 \in B$ → (F)
c) $(6,2)$, pois $6 \in A$ e $2 \notin B$ → (F)
d) $(0,7)$ pois $0 \in A$ e $7 \in B$ → (V)
e) $(5,7)$ pois $5 \notin A$ e $4 \notin B$ → (V)

2) $f(x)=3x^2-4x-7$

a) $f(2)=3.2^2-4.2-7$
$12-8-7$
$-15+12$
$\boxed{-3}$

b) $f(-1)=3.(-1)^2-4.(-1)-7$
$3.1+4-7$
$3+4-7$
$\boxed{0}$

c) Os valores de x para os quais $f(x)=0$
$3x^2-4x-7$

$\Delta = b^2-4ac$
$\Delta = -4^2-4.3.-7$
$\Delta = 16+84$
$\Delta = 100$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{+4 \pm 10}{6}$ $x' = \frac{14}{6} = \boxed{\frac{7}{3}}$ $x'' = \frac{-6}{6} = \boxed{-1}$

d) $\frac{f(x+h)-f(x)}{h}$

$3(x+h)^2-4(x+h)-7-3x^2+4x+7$
$3(x^2+2xh+h^2)-4x-4h-7-3x^2+4x+7$
$3x^2+6xh+3h^2-4h-3x^2$
$6xh+3h^2-4h$
h
$\frac{h(6x+3h-4)}{h} = \boxed{6x+3h-4}$

Lista 1 - Funções e Relações

Maria Eduarda Pabre

26/04/2008

3) $f(x) = x^2 + x - 1$

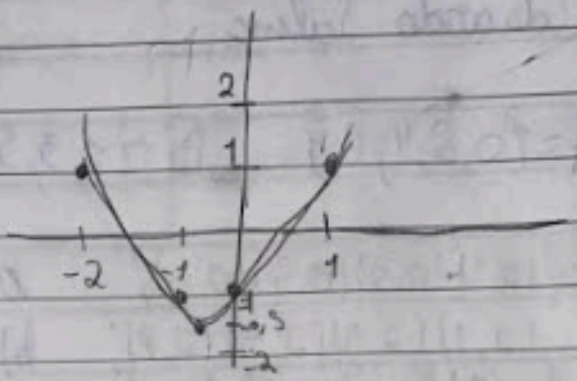
$f(-2) = 4 - 2 - 1 = 1 \rightarrow (-2, 1)$

$f(-1) = 1 - 1 - 1 = -1 \rightarrow (-1, -1)$

$f(-0,5) = 0,25 - 0,5 - 1 = -1,25 \rightarrow (-0,5; -1,25)$

$f(0) = (0, -1)$

$f(1) = (1, 1)$



4) $f(x) = x^2 + 1$

$f(-3) = 10 \rightarrow (-3, 10)$

$f(-2) = 5 \rightarrow (-2, 5)$

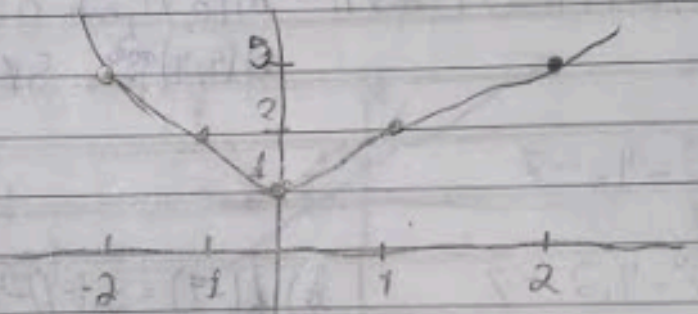
$f(-1) = 2 \rightarrow (-1, 2)$

$f(0) = 0 \rightarrow (0, 1)$

$f(1) = 2 \rightarrow (1, 2)$

$f(2) = 5 \rightarrow (2, 5)$

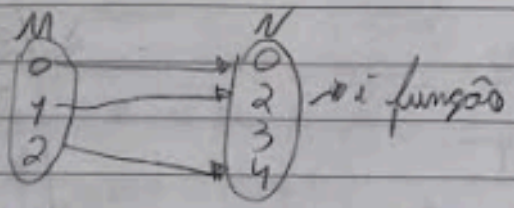
$f(3) = 10 \rightarrow (3, 10)$



5) $R = \{(x, y) \in M \times N \mid y = 2x\}$ · $M = \{0, 1, 2\}$ · $N = \{0, 2, 3, 4\}$

$M \times N = \{(0,0); (0,2); (0,3); (0,4) \ldots (a, 4)\}$

$R = \{(0,0); (1,2); (2,4)\}$

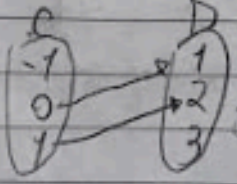


→ é função

6) $R = \{(x, y) \in C \times D \mid y = x + 1\}$ · $C = \{-1, 0, 1\}$ · $D = \{1, 2, 3\}$

$C \times D = \{(-1, 1); (-1, 2) \ldots (1, 3)\}$

$R = \{(0,1); (1,2)\}$



→ Não é função

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Lista 1 - Funções e Relações

Maria Eduarda Palma

26/04/2018

7 - $R = \{(x,y) \in A \times B \mid x < y\}$ $A = \{1,2,4\}$ e $B = \{2,3\}$

$A \times B = \{(1,2);(1,3);\ldots(4,3)\}$

$R = \{(1,2);(1,3);(2,3)\}$

